

Domino Neuron Relay

Make a nerve signal survive a gap.

Win condition: Fastest reliable signal with best explanation.

THE SCIENCE YOU ARE USING

Signals travel in a chain. A line of falling dominoes is a limited model of signal propagation along one axon: each tile triggers the next, while no single tile travels down the track. Real nerve impulses use moving ions, not falling cells, so the model has limits.

Gaps and insulation. A large gap breaks this mechanical relay. In a real neuron, myelin insulates sections of the axon so an impulse travels quickly between exposed nodes. A synapse is a different junction between cells; myelin does not bridge it.

HOW TO BUILD AND RUN IT

- 01 Build a baseline track.** Tape paper down first so tiles cannot slide back. Set dominoes evenly with each gap smaller than a tile is tall (about half a tile), then time one full run.
- 02 Introduce a gap.** Open a larger gap and see whether the signal still crosses.
- 03 Bridge the gap.** Add a bridge piece so this mechanical relay survives. Call it an engineering repair, not a literal synapse or myelin sheath.
- 04 Test a faster relay analogy.** Tape a few tiles into rigid blocks and compare the travel time with the evenly spaced line. Treat fewer active handoffs as an analogy for faster transmission, not a physical model of myelin.
- 05 Explain the analogy.** Explain what the domino relay can represent, and state that myelin insulates one axon while a synapse connects different cells.

Safety: Low risk. Allergy: None.

MATERIALS

Dominoes for one shared backup track	about 16 tiles total; teams rotate
Shared rulers	shared at the backup track
Tape	shared
Index cards	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	100